

Quantitative Methods 3, ZHAW

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Preamble

This script is a continuation of Quantitative Methods 1 and 2 at ZHAW.

In the second part, we learned about the basics of statistical modeling, specifically, using Bayesian and Frequentist approaches.

In this script, we will continue with Reliability, Validity, do a short introduction into artificial intelligence (AI) and also continue to expand our classical statistical toolbox.

This is a first draft as you are the first group to be working with it.

Please feel free to send me suggestions for improvements or corrections.

This **should be a collaborative effort** and will (hopefully) never be finished as our insights on how to present which topic grows over time.

The script can also be seen as a pointer to great sources which are suited to deepen your understanding of the topics. Knowledge is decentralized, and there are many great resources out there - in books, on websites and on YouTube.

For the working setup with R, please see this and the following sections in the first script.

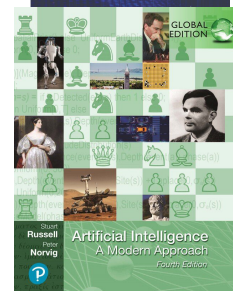
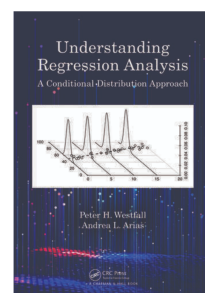
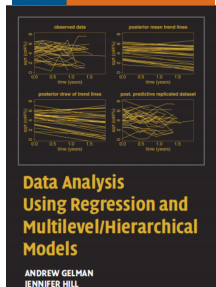
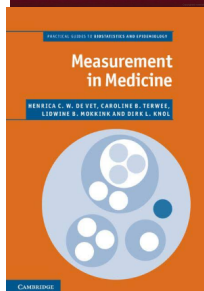
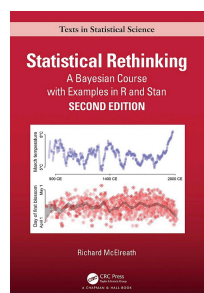
The complete code for this script can be found here. Feel free to copy and modify it for your own purposes.

Separate PDFs of this script (might not be up to date)

- Chapter Reliability and Validity
- Chapter Further Regression Methods

Books we will borrow from are:

- (older online version is Free; current version in ZHAW Library) Statistical Rethinking, YouTube-Playlist: Statistical Rethinking 2023
- (Free) Understanding Regression Analysis: A Conditional Distribution Approach
- Measurement in Medicine
- (3rd edition is free) Artificial Intelligence A Modern Approach
- Data Analysis Using Regression and Multilevel/Hierarchical Models



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consider are opportunity costs of not working. A comparison with both, a foreign education or opportunity costs, justifies the investment in good books. Or: The costs of all the good books of your education combined are probably less than an iPhone Pro.

